INCH-POUND

MS51841E
w/AMENDMENT 1
10 July 2019
SUPERSEDING
MS51841E
10 March 2017

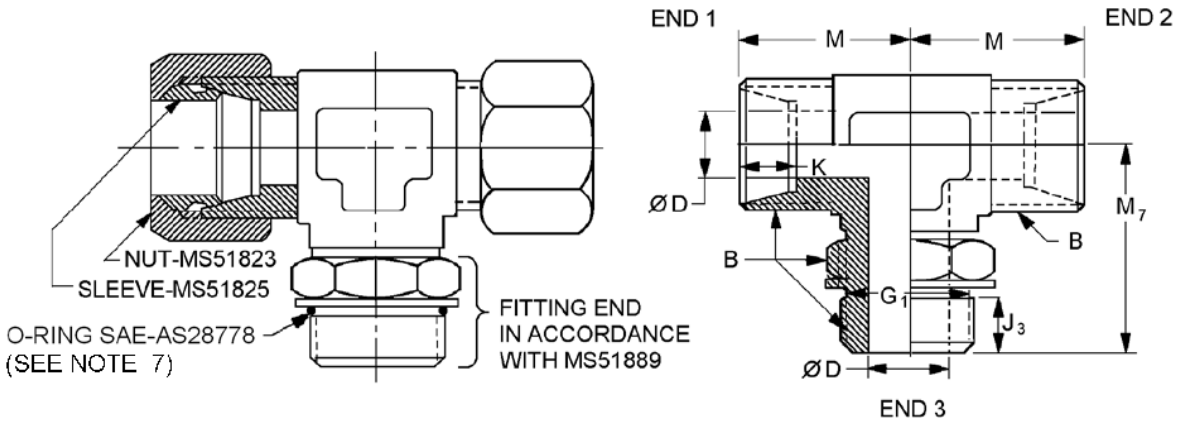
DETAIL SPECIFICATION SHEET

# TEE, TUBE TO BOSS, BOSS ON BENCH, O-RING, FLARELESS TYPE, HYDRAULIC

Reactivated after 10 March 2017 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.




See notes at end of figure.

FIGURE 1. Tee tube to boss.

AMSC N/A

FSC 4730

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| Dash number (see note 5) | | Tube OD Nom inches (mm) | B Straight thread Class 2A | D Diameter Drill inches (mm) | $G_1$ Diameter inches (mm) +.002 (0.05) -.003 (0.08) | $J_3$ Full thread ±.005 (0.13) |
|---|---|---|---|---|---|---|
| Assembly | Body | | | | | |
| -1 | -21 | .1250 (3.175) | .3125 - 24UNF | .093 (2.36) | .250 (6.35) | .234 (5.94) |
| -2 | -22 | .1875 (4.763) | .3750 - 24UNF | .125 (3.18) | .313 (7.95) | .234 (5.94) |
| -3 | -23 | .2500 (6.350) | .4375 - 20UNF | .203 (5.16) | .364 (9.25) | .265 (6.73) |
| -4 | -24 | .3125 (7.936) | .5000 - 20UNF | .234 (5.94) | .427 (10.85) | .265 (6.73) |
| -5 | -25 | .3750 (9.525) | .5625 - 18UNF | .281 (7.14) | .482 (12.24) | .312 (7.92) |
| -6 | -26 | .5000 (12.700) | .7500 - 16UNF | .422 (10.72) | .660 (16.76) | .344 (8.74) |
| -7 | -27 | .6250 (15.875) | .8750 - 14UNF | .500 (12.70) | .773 (19.63) | .391 (9.93) |
| -8 | -28 | .7500 (19.050) | 1.0625 - 12UN | .656 (16.66) | .945 (24.00) | .469 (11.91) |
| -9 | -29 | .8750 (22.225) | 1.1875 - 12UN | .719 (18.26) | 1.070 (27.18) | .469 (11.91) |
| -10 | -30 | 1.0000 (25.400) | 1.3125 - 12UN | .875 (22.23) | 1.195 (30.35) | .469 (11.91) |
| -11 | -31 | 1.2500 (31.750) | 1.6250 - 12UN | 1.093 (27.76) | 1.507 (38.28) | .469 (11.91) |
| -12 | -32 | 1.5000 (38.100) | 1.8750 - 12UN | 1.344 (34.14) | 1.756 (44.60) | .469 (11.91) |
| -13 | -33 | 2.0000 (50.800) | 2.5000 - 12UN | 1.813 (46.05) | 2.381 (60.48) | .469 (11.91) |

| Dash number (see note 5) | | K inches (mm) +.016 (0.41) -.005 (0.13) | M inches (mm) ±.03 (0.8) | $M_7$ inches (mm) ±.03 (0.8) |
|---|---|---|---|---|
| Assembly | Body | | | |
| -1 | -21 | .188 (4.78) | .77 (19.56) | .91 (23.11) |
| -2 | -22 | .234 (5.94) | .83 (21.08) | .94 (23.88) |
| -3 | -23 | .234 (5.94) | .89 (22.61) | 1.03 (26.16) |
| -4 | -24 | .250 (6.35) | .95 (24.13) | 1.13 (28.70) |
| -5 | -25 | .250 (6.35) | 1.05 (26.67) | 1.25 (31.75) |
| -6 | -26 | .305 (7.75) | 1.25 (31.75) | 1.45 (36.83) |
| -7 | -27 | .350 (8.89) | 1.42 (36.07) | 1.70 (43.18) |
| -8 | -28 | .350 (8.89) | 1.58 (40.13) | 1.94 (49.28) |
| -9 | -29 | .350 (8.89) | 1.66 (42.16) | 2.00 (50.80) |
| -10 | -30 | .415 (10.54) | 1.73 (43.94) | 2.05 (52.07) |
| -11 | -31 | .415 (10.54) | 1.89 (48.01) | 2.25 (57.15) |
| -12 | -32 | .485 (12.32) | 2.02 (51.31) | 2.39 (60.71) |
| -13 | -33 | .485 (12.32) | 2.45 (62.23) | 2.89 (73.41) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514.
5. Assemblies are only furnished to this specification (body, nut, and sleeve). Bodies are not to be stocked stored or issued.
6. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.
7. Procure the O-ring separately and install immediately before installation of the fitting.

FIGURE 1. Tee, tube to boss - Continued.

REQUIREMENTS:



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Tee, tube to boss shall be as specified on figure 1 and in table I.

Materials shall be in accordance with MIL-DTL-18866.

Finish. Finishes shall be as specified in table I. All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

TABLE I. Material and finish identification codes.

| PIN code material/plating finish | Material | Plating finish |
|---|---|---|
| Blank | Steel | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/ 5/ 7/ |
| H | Steel | Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 2/ |
| J | Steel | Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/ 3/ |
| M | Nickel-copper alloy UNS N04400 | No additional finish. |
| N | High-chromium nickel alloy UNS N06690 | No additional finish. |
| P | Steel | Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 4/ |
| R | Steel | Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 4/ 5/ |
| SS | Corrosion resistant steel | No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7, free machining alloy UNS S30300 method 1, type 2 or 4. |
| T | Titanium | Anodize in accordance with SAE-AMS2488 type 2. 6/ 8/ |
| TF | Titanium | Fluoride phosphate in accordance with SAE-AMS2486. 6/ 8/ |
| Z | Steel | Zinc plating in accordance with ASTM B695, type II, class 8. 4/ 5/ |
| ZC | Steel | Zinc may be any zinc plating from PIN codes J, R, and Z with a colored chromate coating. 4/ 5/ |

1/ Embrittlement test need not be run.
2/ Hexavalent chromium free. Finish shall be RoHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
4/ Not for use in aircraft. Requires approval from the Program Office for all applications. Zinc-plated materials should be avoided in hydraulic systems with water-glycol fluids. Reference ASTM MNL37 - Fuels and Lubricants Handbook.
5/ These finishes have a chromate conversion coating which includes hexavalent chromium and are not recommended for Army ground systems. (Notes contined on next page)
6/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293.



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7/ Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.

8/ Titanium shall not be used in oxygen or potable water systems.

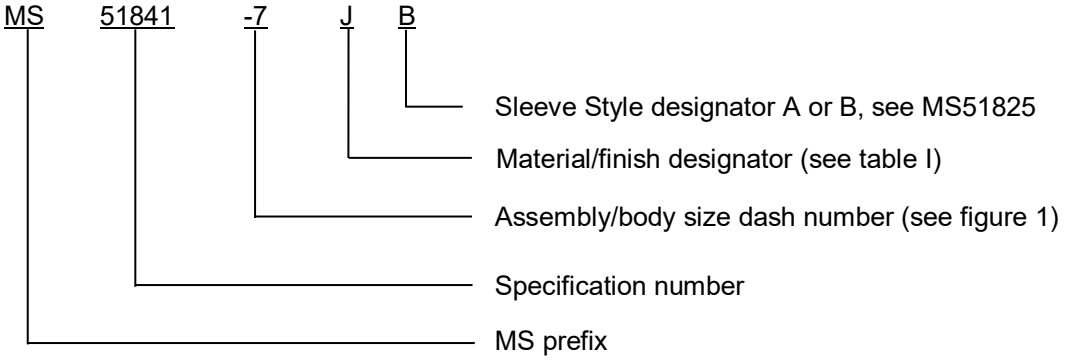
Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using one of the brush plating processes below as appropriate to primary finish. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS-2451/5.

b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS-2451/9.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.

PIN: The PIN consists of the letters “MS”, the specification number, a dash, number for a tee, tube to boss assembly/body size, and a letter for material finish designator.




PIN example: MS51841-7JB indicates a tee, tube to boss, ends 1 and 2 are .6250 inch (15.875 mm) tube OD, end 3 is a .8750 - 14UNF-2A thread, Zinc-Nickel plated, and B style sleeve.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.

Class I and II ozone depleting substances (ODS) shall not be used in MS51841 or any referenced procedures.

Tapered pipe thread connections are generally prohibited in high shock applications.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.



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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

| | | |
|---|---|---|
| MIL-DTL-16232 | SAE-AMS-STD-595/36076 | SAE-AMS-STD-595/36270 |
| MS51823 | SAE-AMS-STD-595/36081 | SAE-AMS-STD-595/36280 |
| MS51825 | SAE-AMS-STD-595/36099 | SAE-AMS-STD-595/36293 |
| MS51889 | SAE-AMS-STD-595/36118 | SAE-AMS2417 |
| ASTM B117 | SAE-AMS-STD-595/36134 | SAE-AMS2451/5 |
| ASTM B633 | SAE-AMS-STD-595/36152 | SAE-AMS2451/9 |
| ASTM B695 | SAE-AMS-STD-595/36170 | SAE-AMS2486 |
| ASTM F1136/F1136M | SAE-AMS-STD-595/36173 | SAE-AMS2488 |
| ASTM MNL37 | SAE-AMS-STD-595/36176 | SAE-AMS2700 |
| SAE-AMS-C-81562 | SAE-AMS-STD-595/36231 | SAE-AS28778 |
| SAE-AMS-QQ-P-416 | SAE-AMS-STD-595/36251 | SAE-J514 |
| | | T9070-AL-DPC-020/077-2 |

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2019-051)

Review activities:
Army - AR, MI
Navy - CG, SA
Air Force - 71

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